# Freshness and Independent Supervision

## An operating-systems study on a heterogeneous edge rig

Raspberry Pi 5 / Ubuntu 24.04 + STM32F446RE / FreeRTOS

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Defense evidence snapshot: 13 October 2026

# 1. Motivation

Artificial intelligence (AI) is a variable Linux workload.

- A late result can be correct but unusable.
- A live process can keep sending stale results.
- A separate supervisor can reject stale proposals.

Figure 1 locates the consumer's freshness boundary.

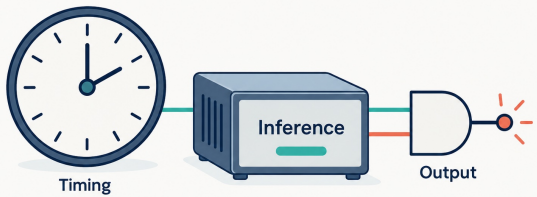



Figure 1: The consumer needs timely information.

Study when information remains useful, alongside how fast work completes.

# 2. Problem and research questions

- **RQ1:** How does contention change Linux timing tails?
- **RQ2:** Where should a latest-value policy discard old work?
- **RQ3:** Can the microcontroller (MCU) distinguish silence from stale results?

**Metrics:** wake-up latency, input-to-reception delay,
Age of Information (AoI), and reason-coded state transitions.

> The contribution is an OS experiment and evidence trail; AI supplies load.

# 3. System architecture

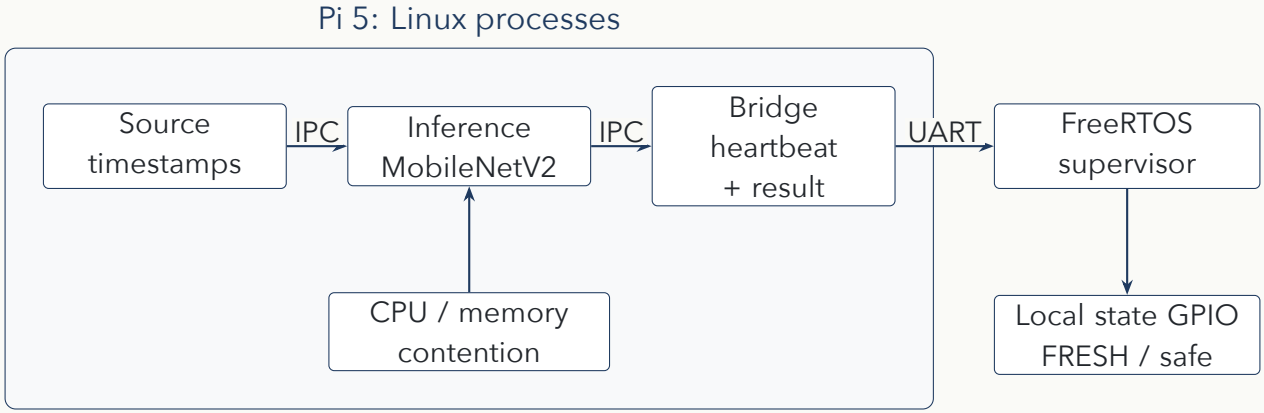



Figure 2: A UART boundary separates Linux work from MCU judgment.

Figure 2 separates inter-process communication (IPC) from the UART serial link.

The STM32 supervises freshness independently of Linux scheduling.

# 4. Scheduling and contention profiles

### Applied software-campaign profiles

- P0 source/bridge: OTHER, CPU 1, unlocked.
- P1 source/bridge: FIFO 60/70, CPU 1, memory locked.
- Both profiles: inference OTHER, CPU 0, unlocked.

### Mechanisms and scope

CPU (central processing unit) affinity controls placement.

`mlockall` locks resident memory.

Control groups (cgroups) can cap CPU service.

**PLANNED:** quota and packaged P3 kernel comparisons; no measured benefit.

[1] Linux scheduling. Applied E2 settings change priority and locking together. Cyclictest P1 is FIFO 50, a separate configuration. Loads and stressed cores: method, page 9.

> CPU priority and affinity do not isolate shared memory bandwidth.

# 5. IPC and buffering policy

Unix datagram sockets separate source, inference and bridge processes.




Figure 3: Replacement acts only at its chosen waiting queue.

Figure 3: bridge FIFO holds up to 256 results; latest keeps one unsent result.

Input latest drains waiting inputs before inference; it does not preempt current work.

Replacing after expensive work cannot remove a backlog before that work.

# 6. Clocks and freshness

**One Pi clock per measurement**

`CLOCK_MONOTONIC`: release deadlines.

`CLOCK_MONOTONIC_RAW`: recorded timestamps and age-at-send.

**Do not mix those timestamp domains.**

[2] Linux clock semantics.

**Independent MCU clock**

$A_s = t_{send} - t_{input}$ on the Pi.

$\widehat{A} = A_s + (t_{MCU} - t_{receipt}) + B$.

$B = 5$ ms: configured transit allowance.

Echo exchange fits offset and drift [3].

Mapping uncertainty remains; no proven transit bound.

AoI is the age of the newest useful input at the consumer.

# 7. Independent supervisor

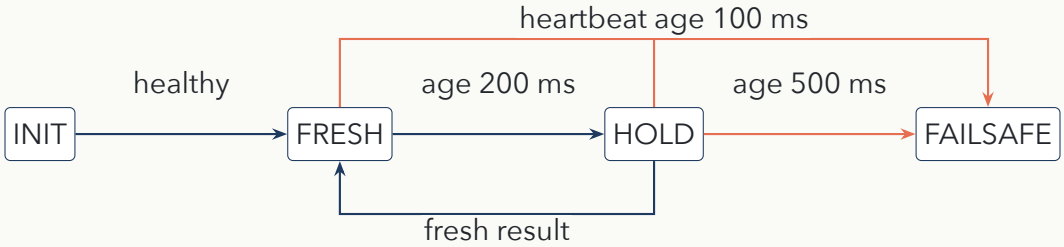



Figure 4: Separate timers test liveness and source freshness.

Figure 4 uses configured, provisional thresholds.

FAILSAFE is latched. Portable core supports local healthy rearm (1 s); this port uses board reset.

FreeRTOS fixed-priority tasks [4]; single supervisor owner, static resources. No independent MCU hardware watchdog.

> A continuing heartbeat cannot make an old result fresh.

# 8. Experimental method

**Retained rig**

Pi 5, 4 GB; Ubuntu 24.04.
Stock `6.8.0-1064-raspi` kernel.

STM32F446RE; FreeRTOS.
USART1, 921600 baud; direct memory access (DMA) reception.

3.3 V UART: Pi TX/RX ↔ PA10/PA9; shared ground.

**Load key (stress-ng on the Pi)**

CPU: compute workers, CPUs 2/3.
Stream: streaming arrays, CPU 2.
Cache: 2 MiB working set, CPU 2.
Memrate: read/write traffic, CPUs 2/3.

Timer: CPU 0; pipeline inference: CPU 0; source/bridge: CPU 1.

Cyclictest: 3×120 s/cell; E1–E3: 3×60 s/cell, pseudo-terminal (PTY).
Inference: 50 warm-ups + 1,000 calls/run; analyzer: 15 fault captures.

Performance governor: timer/pipeline; ondemand: model-only baseline (CPUs 0–3). PTY excludes physical UART/MCU. Exact load commands: CLAIMS.md.

> Keep timer, model, software pipeline and physical-fault scopes separate.

# 9. Run validity and rule versions

| Evidence | Stored v1 acceptance | Stricter v2 replay |
|---|---|---|
| Cyclictest | 30/30 | 6/30: idle only |
| E1–E3, PTY | 45/45 | 0/45: RAW monitor times absent |
| Original E4 entries | 17/25 | 17/25: unchanged |

V2 requires whole-window RAW power timestamps, stressor liveness, and clock fits no older than 5 s when mapping MCU time.

MEASURED v1 observations remain reported with their evidence limitation. Replay rejection is missing proof, not proof that every old run malfunctioned. [5] Results inventory.

A measured number and a fully qualified comparison are different claims.

# 10. Timer wake-up under contention

**MEASURED – v1; v2 accepts idle runs only**

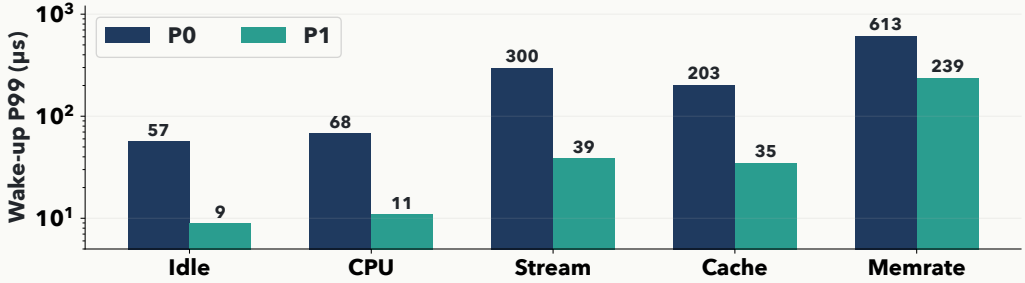



Figure 5: Median run P99; 3 runs/cell, locked memory, CPU 0.

Figure 5: FIFO P99 rises from 9 to 239 $\mu$s under memrate.

P0 OTHER / P1 FIFO 50. P50 gap includes timer slack; stressed v2 replay lacks recorded stressor liveness.

Priority reduces CPU tails here; memory-contention tails remain.

# 11. Inference baseline

**MEASURED – accepted subset of a mixed collection**

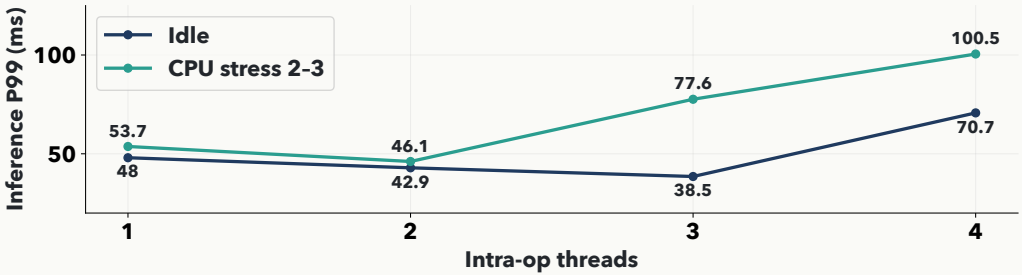



Figure 6: MobileNetV2 FP32, ONNX Runtime; inference only.

Figure 6: idle scaling does not predict loaded tails.

21/24 accepted; idle reps 3/3/3/1, loaded 3/3/3/2. Three 4-thread runs excluded for live under-voltage. Ondemand; affinity CPUs 0–3.

More model threads can compete with the stressed cores.

# 12. E1: contention reaches the data path

**MEASURED – PTY, v1; 15/15 stored, 0/15 under v2 replay**

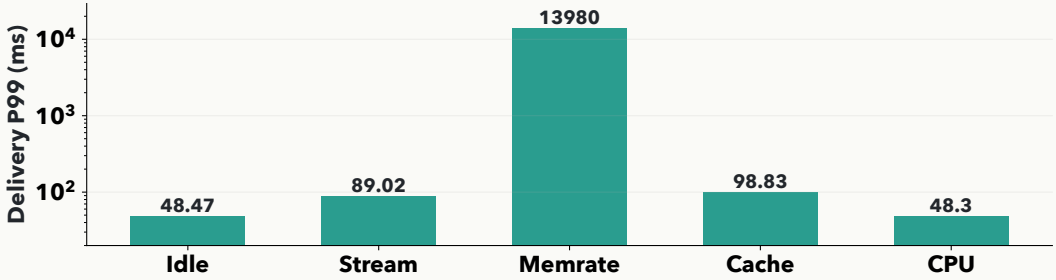



Figure 7: Input-to-Pi-reception P99; median across 3 runs.

Figure 7: memrate reaches 13,980 ms; idle is 48.47 ms.

P0, performance governor. Service includes upstream waiting and pre/postprocessing; this is not model execution time. Logarithmic axis.

The memory-loaded pipeline accumulates seconds of waiting.

# 13. E2: runtime tuning has limits

**MEASURED – PTY, v1; 18/18 stored, 0/18 under v2 replay**

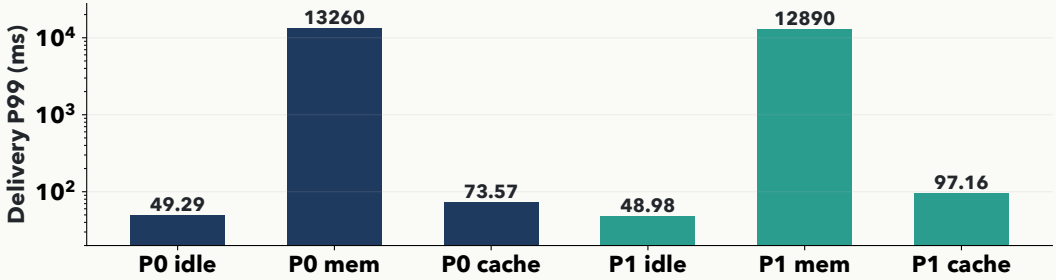



Figure 8: Matched loads; P99 of delivery, median across 3 runs.

Figure 8: memrate P99 stays 13,260 / 12,890 ms for P0 / P1.

P0 source/bridge: CPU 1, unlocked. P1: FIFO 60/70, CPU 1, locked. Both: inference OTHER, CPU 0, unlocked. This compares profiles, not scheduling class alone.

P1 does not remove memory-loaded backlog or improve every cache result.

# 14. E3: queue observations and interpretation

**MEASURED – PTY, v1; 12/12 stored, 0/12 under v2 replay**

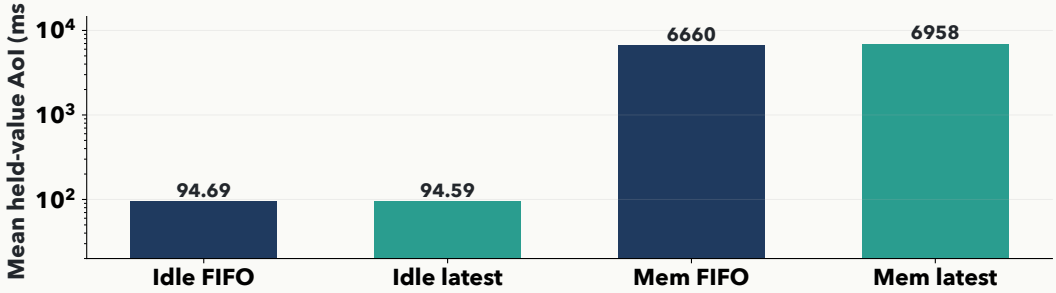



Figure 9: Initialized time-average AoI; stalls and final tail included.

Figure 9: loaded bridge FIFO/latest average ages are 6,660 / 6,958 ms.

MEASURED v1: zero bridge replacements; 100 ms inputs; 122–130 ms completion intervals. FLAGGED E3b: input-latest P99 254–268 ms vs FIFO 13.8–15.9 s; all loaded runs power-invalid.

INFERRED: backlog before inference. Clean input-policy benefit: UNKNOWN.

# 15. E4: interpret the transition reason

**MEASURED – accepted MCU transitions; short-window scopes differ**

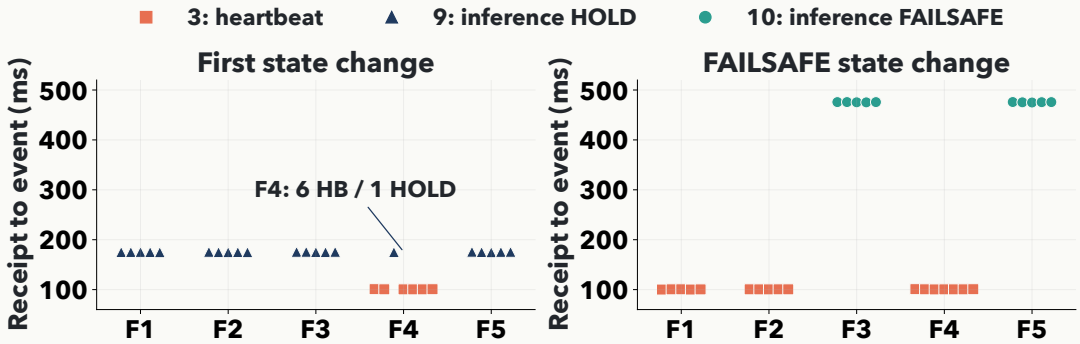



Figure 10: One point per accepted run; horizontal offsets separate close samples.

Figure 10: reason 3 uses heartbeat; reasons 9/10 use inference.

F1 kill, F2 bridge stop, F3 inference stop, F4 FIFO-99 hog, F5 inference suppression. Accepted F1/F2/F3/F5: 5 each; F4: 7/14. F4 first events mix reasons 9 and 3.

About 175/475 ms after receipt matches 200/500 ms source-age thresholds.

# 16. Independent analyzer cross-check

**MEASURED – retained physical captures; collection remains FLAGGED**

$\Delta = L - M$; positive means the analyzer interval is longer.
$L$: last physical UART byte end → both PA5/PB5 low.
$M$: last accepted heartbeat receipt → FAILSAFE STATE_EVENT time.

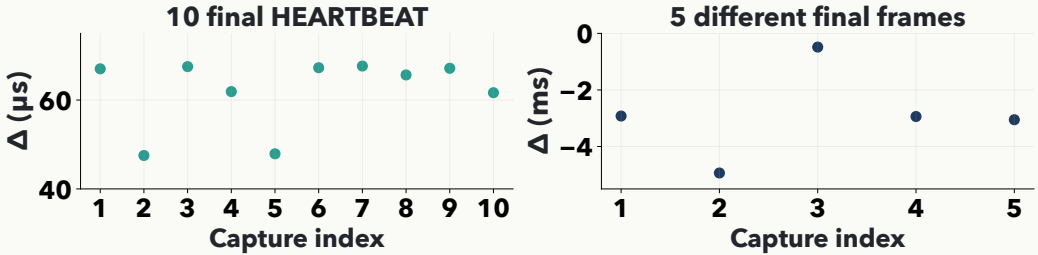



Figure 11: Residuals by capture index; full IDs in qa/LA_capture_key.md.

Figure 11: 10/15 final HEARTBEAT references give +47.5–67.6 $\mu$s.

Residual includes timestamp placement, reception/output delay and clock effects. Other 5 end with INFERENCE/ECHO. 8 MHz sampling; absolute calibration UNKNOWN.

Matching frame types aid comparison; they do not qualify the FLAGGED collection.

# 17. Limitations

- 5 V / 3 A supply; live under-voltage excludes loaded runs.
- Original real-UART E3: 18/18 FLAGGED; no clean physical AoI.
- V2 replay exposes missing monitor timestamps and stressor proof.
- Small samples and observed maxima are not worst-case bounds.
- Firmware review fixes compile; they are **not re-flashed or rig-verified**.

Analyzer captures add a GPIO observation to the older stage report; they do not close transit, oscillator, rearm or watchdog qualification.

> The evidence supports this experimental envelope, with explicit exclusions.

# 18. Conclusion and contributions

- **Timing:** priority improves observed timer tails; shared contention persists.
- **Freshness:** upstream backlog is INFERRED from measured queue behavior.
- **Supervision:** independent MCU timers separate stale work from silence.
- **Artifact:** shared protocol, instrumented pipeline, raw data and exclusions.

Clean input-policy benefit remains **UNKNOWN**; loaded E3b is **FLAGGED**.

The teaching result connects scheduling, IPC, clocks and real-time state decisions.

> Defensible measurements include the failures and the limits of each claim.

# 19. Future work

**PLANNED**

- Restore clean power; rerun input-policy and physical E3 under v2.
- Flash reviewed firmware; verify DMA, stack margins and event logging.
- Match analyzer references; calibrate clocks, transit and rearm margins.
- Extend fault exposure and recovery; reach five valid repeats per cell.

Quota and packaged P3 kernel comparisons remain optional matched experiments.

> The next deliverable is qualified physical freshness evidence.

# 20. References


1. Linux man-pages. *sched(7)*: scheduling policies and affinity.
2. Linux man-pages. *clock_gettime(2)*: monotonic and raw clocks.
3. D. Mills. *RFC 4330*, January 2006, section 5: four-timestamp math.
4. FreeRTOS. *Tasks and Co-routines*: kernel task model.
5. Project artifact. *STAGE_REPORT.md*, *results/README.md*, retained data (11–12 October 2026); source snapshot `5e3f461`.


Primary documentation checked 13 October 2026. Exact figure inputs and hashes: qa/source_hashes.json. Slide-to-evidence mapping: CLAIMS.md.

# Backup

Questions, evidence and configuration detail

# B1. UART protocol frame

| Field | SOF | Version | Type | Seq | Length | Payload + CRC |
|---|---|---|---|---|---|---|
| Bytes | 2 | 1 | 1 | 2 | 1 | 0–64 + 2 |

- SOF: A5 5A; little-endian fields; version 3 encoder.
- CRC-16/CCITT-FALSE checks version through payload.
- INFERENCE: 35 bytes, including age-at-send.
- HEARTBEAT carries session identity; STATE_EVENT carries reason.

CALCULATED: 35 bytes × 10 bits / 921600 baud = 0.380 ms nominal line time. This excludes all software delay. Source: protocol/PROTOCOL.md.

Integrity, sequence and freshness checks serve different purposes.

# B2. Clock math and measurement boundaries

$T_1, T_4$: Pi; $T_2, T_3$: MCU, in microseconds.

$$O = \frac{(T_2 - T_1) + (T_3 - T_4)}{2}, \qquad d = (T_4 - T_1) - (T_3 - T_2)$$

$$O(t) = O_R + D(t - R), \qquad t_{Pi} = R + \frac{t_{MCU} - R - O_R}{1 + D}$$

- Asymmetric delays bias offset; fit residual is not absolute accuracy.
- V2 rejects fits older than 5 s; accepted RMS ceiling is 200 $\mu$s.
- Held-value AoI integrates age over initialized time, including final tail.

[3] RFC 4330, section 5; protocol/CLOCK_SYNC.md. RMS ceiling is a configured acceptance choice.

# B3. Threshold rationale and rearm

| Timer | Configured | Retained observation |
|---|---|---|
| Heartbeat | 100 ms | F1/F2 reason 3: 100.128–100.641 ms |
| Source HOLD | 200 ms | F5 reason 9: 174.950–175.917 ms after receipt |
| Source FAILSAFE | 500 ms | F5 reason 10: 474.950–475.917 ms after receipt |
| Healthy rearm | 1,000 ms | Core supports it; board port requires reset |
| Transit allowance | 5 ms | Provisional; no validated upper bound |

Result age also includes age-at-send: receipt elapsed is not source age.

Configured values are ASSUMED choices, not calibrated safety budgets. main.c; supervisor.c; raw F5 events. E4 threshold snapshots are empty/null.

Healthy heartbeat alone cannot clear latched FAILSAFE.

# B4. Full run-validity inventory

| Collection | Accepted | Scope / qualification |
|---|---|---|
| Cyclictest | 30/30 v1 | v2: 6/30, idle only |
| Inference baseline | 21/24 | Mixed; three live under-voltage exclusions |
| E1 / E2 / E3 | 15 / 18 / 12 | v1 PTY; v2: 0/45 |
| E3b input policy | 3/12 | All six loaded runs FLAGGED |
| Real-UART E3 (original / rerun) | 0/18 each | All FLAGGED; distinct validity rules |
| Original E4 | 17/25 entries | 20 performed; 5 F5 unperformed |
| Extra F4 / later F5 | 5/9 / 5/5 | Short-window transition checks |
| Logic analyzer | 15 captures | FLAGGED collection; 1 incomplete attempt |

Source: results/README.md. Missing STATE_EVENT alone cannot prove unhealthy steady state. No excluded observation is promoted into a clean comparison.

# B5. Full E4 table: MCU-local timing

**MEASURED – milliseconds; nearest-rank P50 / P95 / max**

| Fault | Valid | First P50 / P95 / max | FAILSAFE P50 / P95 / max |
|---|---|---|---|
| F1 | 5/5 | 175.317 / 175.535 / 175.535 | 100.456 / 100.641 / 100.641 |
| F2 | 5/5 | 175.206 / 175.709 / 175.709 | 100.577 / 100.596 / 100.596 |
| F3 | 5/5 | 175.754 / 175.815 / 175.815 | 475.754 / 475.815 / 475.815 |
| F4 | 7/14 | 100.639 / 175.229 / 175.229 | 100.556 / 100.931 / 100.931 |
| F5 | 5/5 | 175.596 / 175.917 / 175.917 | 475.596 / 475.917 / 475.917 |

F1/F2: first 9, FAILSAFE 3.

F3/F5: first 9, FAILSAFE 10. F4: six direct 3; one first 9.

F4 combines original 2/5 with extra 5/9; short windows differ. Reasons 9/10 reference INFERENCE; reason 3 references HEARTBEAT. P95 = maximum at these sample sizes.

> A mixed-reference first-transition summary is not heartbeat latency.

# B6. Full E4 table: mapped fault-onset timing

**MEASURED – milliseconds; nearest-rank P50 / P95 / max**

| Fault | Valid | First P50 / P95 / max | FAILSAFE P50 / P95 / max |
|---|---|---|---|
| F1 | 5/5 | 83.342 / 83.561 / 83.561 | 85.357 / 86.209 / 86.209 |
| F2 | 5/5 | 83.151 / 83.715 / 83.715 | 85.715 / 86.113 / 86.113 |
| F3 | 5/5 | 83.646 / 85.914 / 85.914 | 383.620 / 385.888 / 385.888 |
| F4 | 7/14 | 85.625 / 87.871 / 87.871 | 87.277 / 97.344 / 97.344 |
| F5 | 5/5 | 83.608 / 85.563 / 85.563 | 383.603 / 385.558 / 385.558 |

Mapped intervals begin at Pi software fault onset, through the echo estimator.

MCU-local intervals begin at the last accepted message selected by reason.

Accepted runs record zero false transitions in their checked pre-fault windows.

Clock residual and timestamp placement limit mapping accuracy. Extra F4 and F5 do not qualify original recovery/long-exposure gates. No false-trip rate or worst-case bound follows.

# B7. Claim-to-evidence map

| Claim | Evidence label | Retained source |
|---|---|---|
| FIFO timer tails differ | MEASURED v1 | cyclictest summaries |
| Memory-loaded backlog | MEASURED v1, PTY | E1/E2 tables, infer.csv |
| Zero replacements; slow completions | MEASURED v1, PTY | E3 table; infer.csv |
| Backlog before inference | INFERRED | E3 queue observations |
| Clean input-latest benefit | UNKNOWN; E3b FLAGGED | Power-invalid E3b logs |
| Heartbeat / source-age decisions | MEASURED, scoped | E4/F5 STATE_EVENT |
| 10/15 matching analyzer references | CALCULATED; collection FLAGGED | VCD + CRC decode + events |
| Clean physical AoI / safety bound | UNKNOWN | Not established |

Analyzer: calculated comparison of retained MEASURED captures; matching frame types do not establish run validity. Full qualifications: CLAIMS.md; hashes: qa/source_hashes.json.

# B8. Tools and branch map

**Execution environment**

Mac: XeLaTeX, plots, cross-compile, flash.

Pi: Linux pipeline, stress-ng, cyclictest, metadata.

STM32: FreeRTOS tasks, TIM2, DMA UART, state GPIO.

Analysis: e_summary, e4_summary, la_analyze.

**Artifact provenance**

`feat/pi-timing-tools`
`feat/inference-baseline`
`feat/fw-rx-fix`
↓ merged release evidence
`release/os-final`
↓ submission worktree
`feat/os-submission` / `5e3f461`

Branch history is provenance, not a completion claim. This deck build changes only submission/defense; it runs no hardware experiments.

Regenerate this deck with its retained script and source manifest.